D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM MEMBERS

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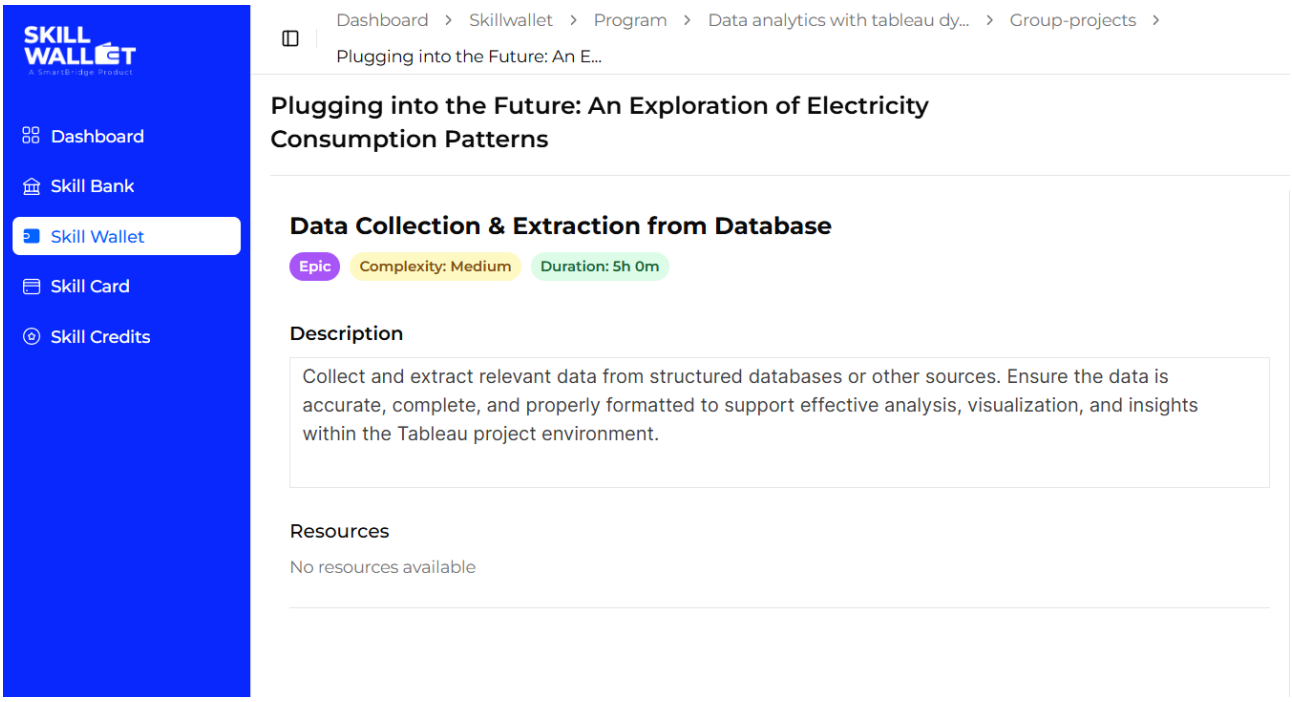
3) **Pranav Dabane**
4) **Prasad Karande**

1) Data Collection & Extraction from Database :

❖ Collect the dataset :

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the given dataset link drive opens , then download the dataset i.e. **Consumption.csv**

❖ Storing Data in DB & Perform SQL Operations :

Step 1: Click on "Microsoft Excel" or "Text File"

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)
Text File (for .csv)
Click the correct option.

Step 2: Select Your File

Browse your computer.
Select your dataset file.
Click Open.

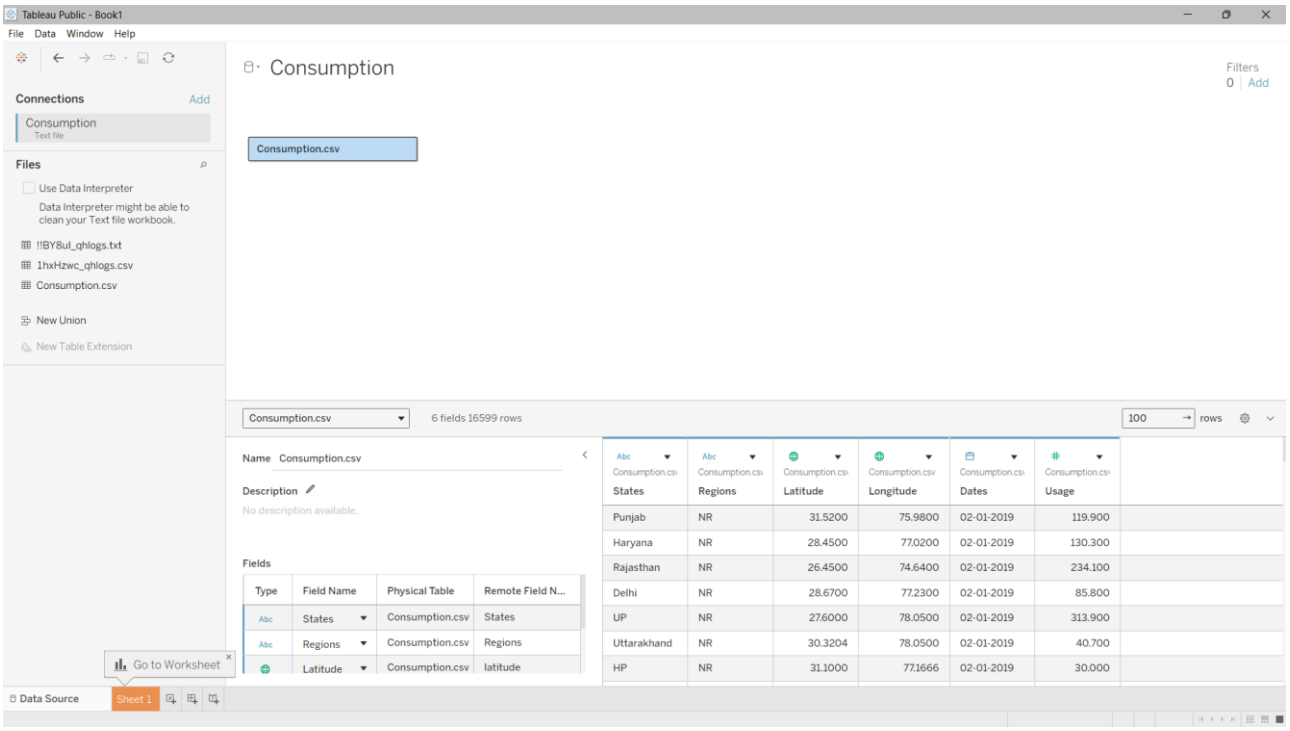
Step 3: Data Source Page

Your sheet names will appear on the left side.
Drag the sheet to the centre area.
Tableau will show a preview of your data.

2) Data Preparation:

❖ Prepare the Data for Visualization :

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



3) Data Visualization:

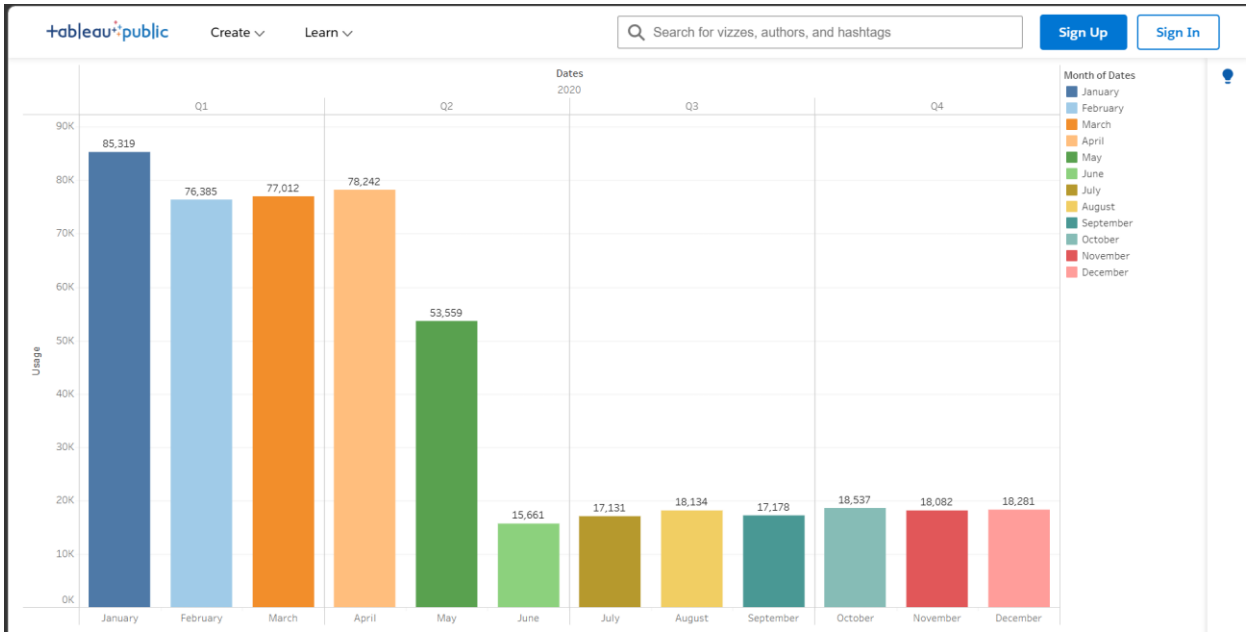
❖ No of Unique Visualizations:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

- **Visualization Part 1 :**

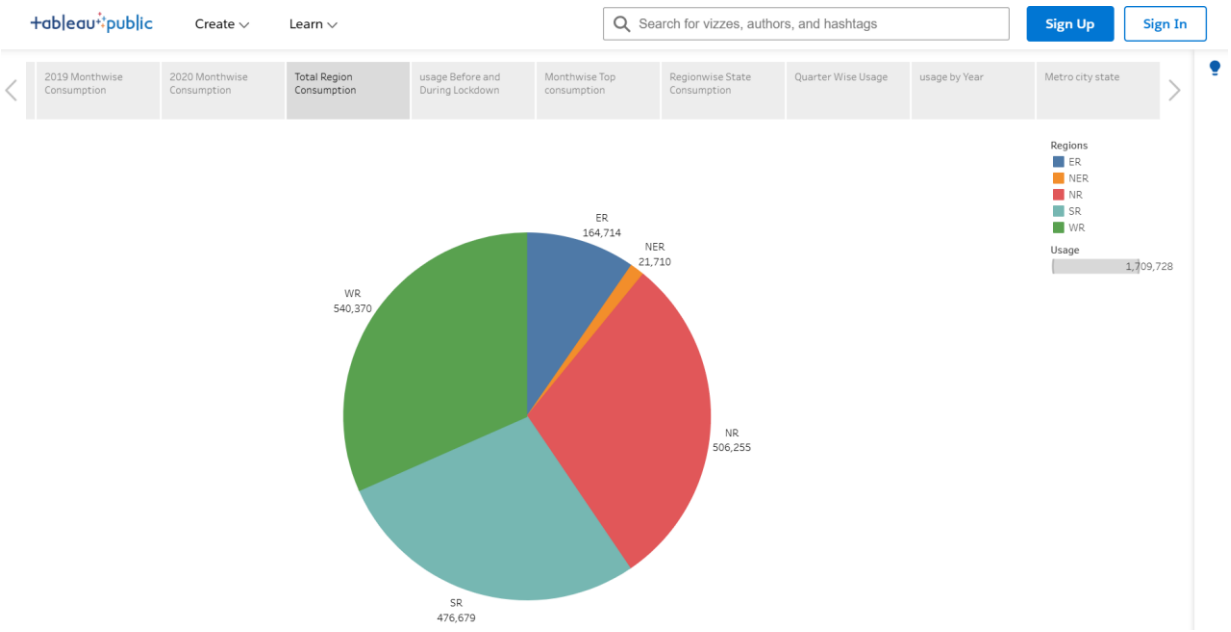
1) To Understand-2019 and 2020 Consumption :

- Drag and Drop **Year** to **Columns**.
- Drag and Drop **Consumption** to **Rows**.
- Change chart type to **Bar Chart**.
- Add **Label** from the Marks card to display values.



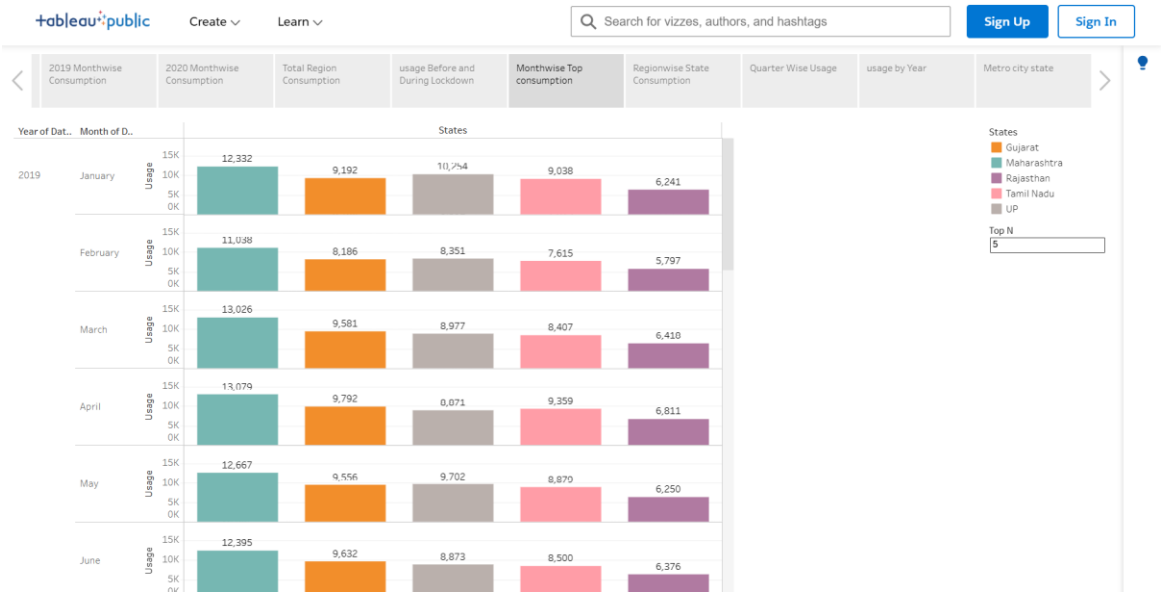
2) Total Consumption :

- Drag **Region** to **Rows**.
- Drag **Consumption** to **Columns**.
- Choose **Bar Chart**.
- Sort descending to show highest consumption first.



3) Top N States:

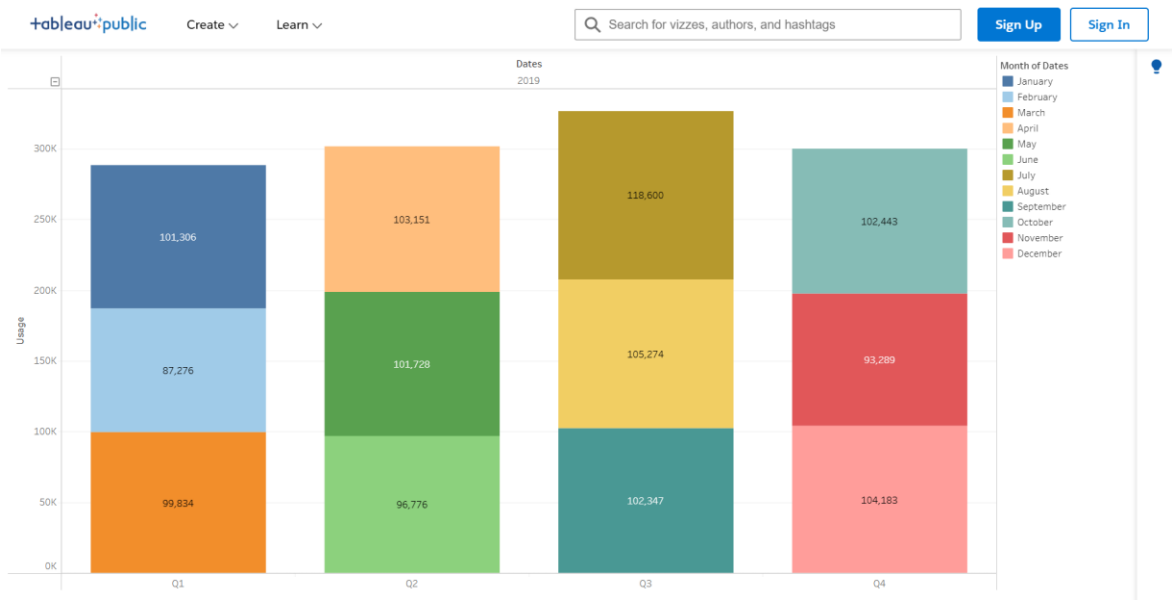
- Drag **Date or Year** to **Columns**.
- Drag **Consumption** to **Rows**.
- Change chart type to **Line Chart**.



- **Visualization Part 2 :**

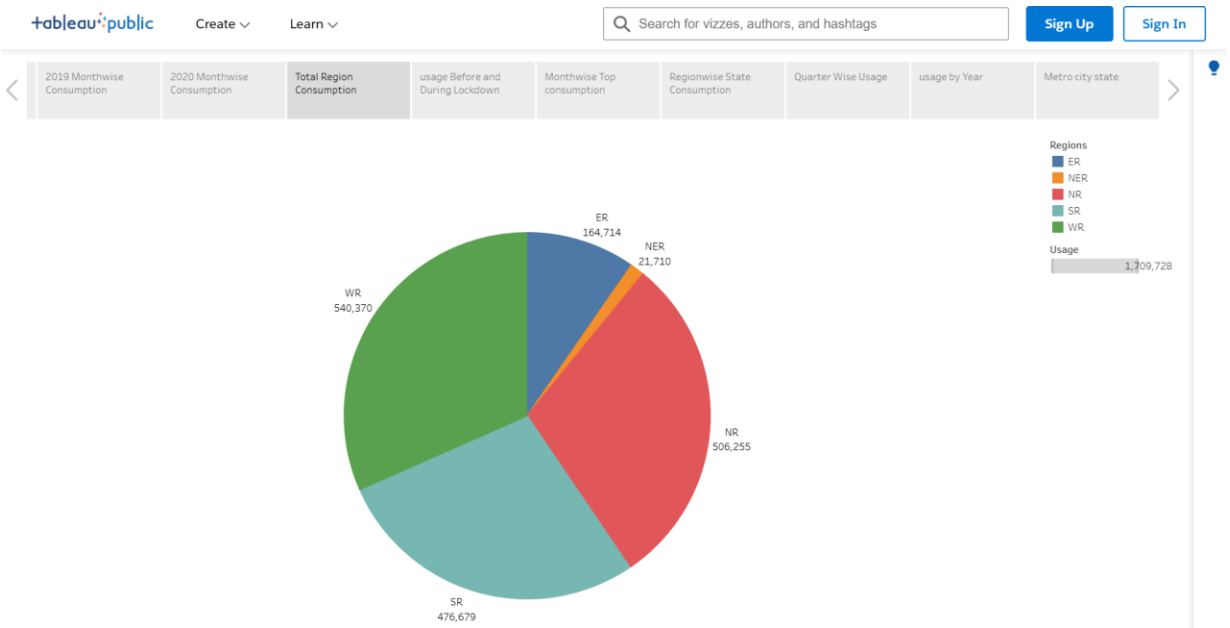
1) To Understand-2019 Month wise Consumption :

- Drag **Date** to **Columns**.
- Click the Date field and change it to **Month**.
- Drag **Consumption** to **Rows**.
- Drag **Year(Date)** to **Color** in the Marks card.
- Change chart type to **Line Chart**.



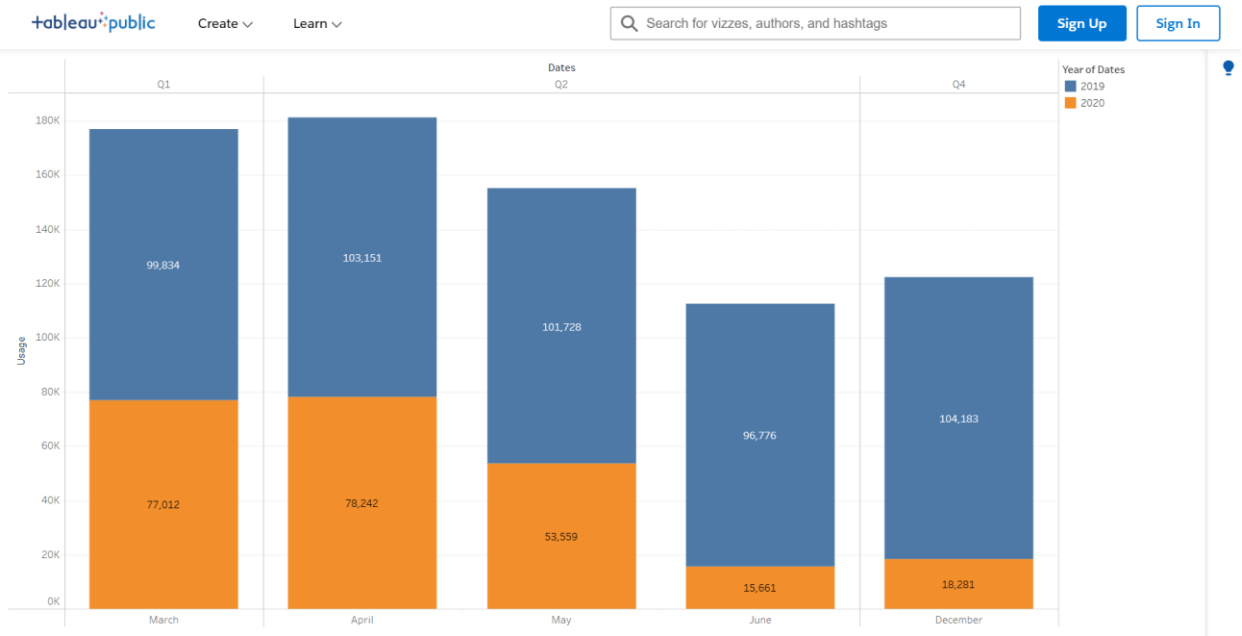
2) Total Consumption by region :

- Drag **Region** to **Rows**.
- Drag **Consumption** to **Columns**.
- Select **Bar Chart** from the Show Me panel.
- Click **Sort Descending**.



3) Usage Before and After Lockdown :

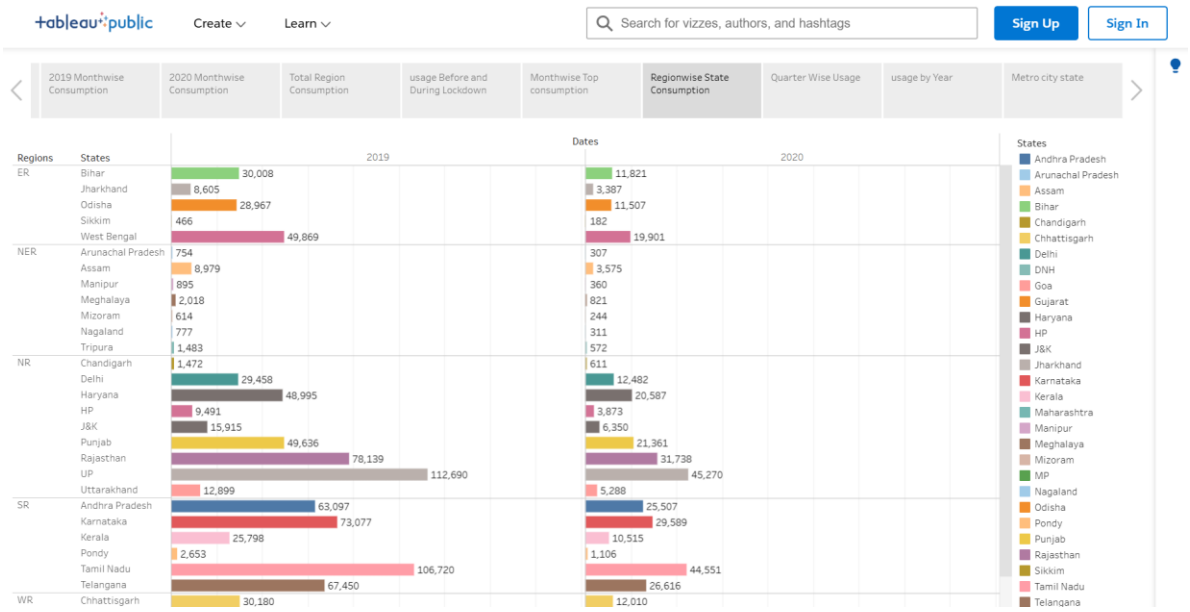
- Click **Analysis → Create Calculated Field**.
- Name it **Lockdown Period**.



- **Visualization Part 3 :**

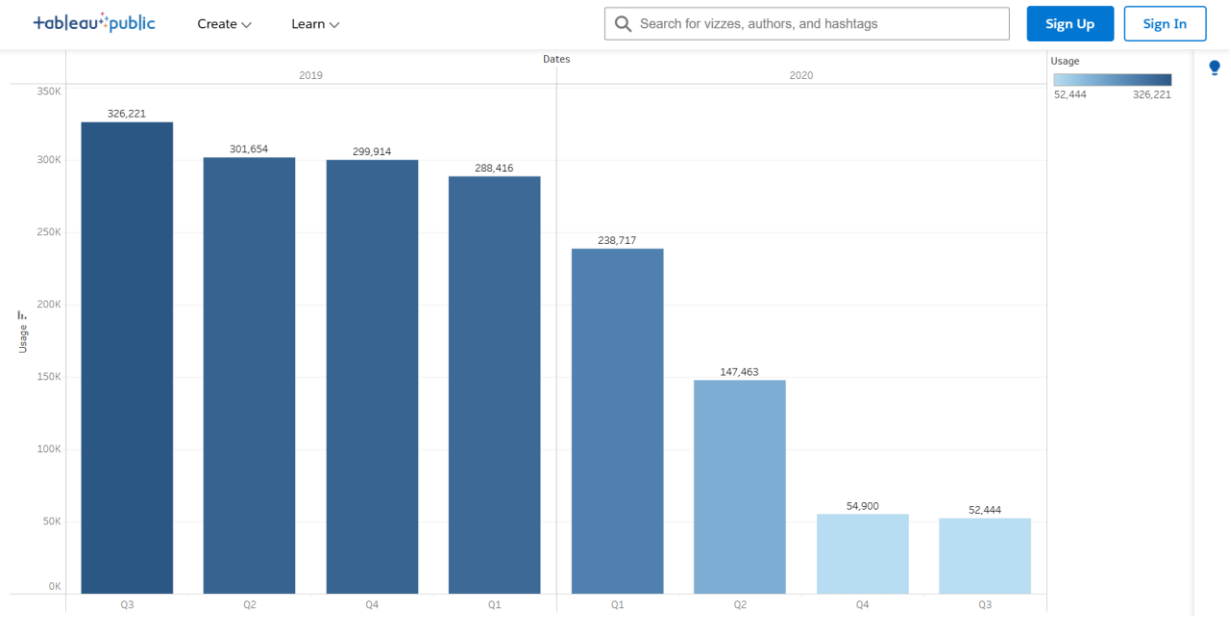
1) State Usage :

- **Drag State → Rows.**
- **Drag Usage / Consumption → Columns.**
- **Select Bar Chart.**
- **Sort descending.**



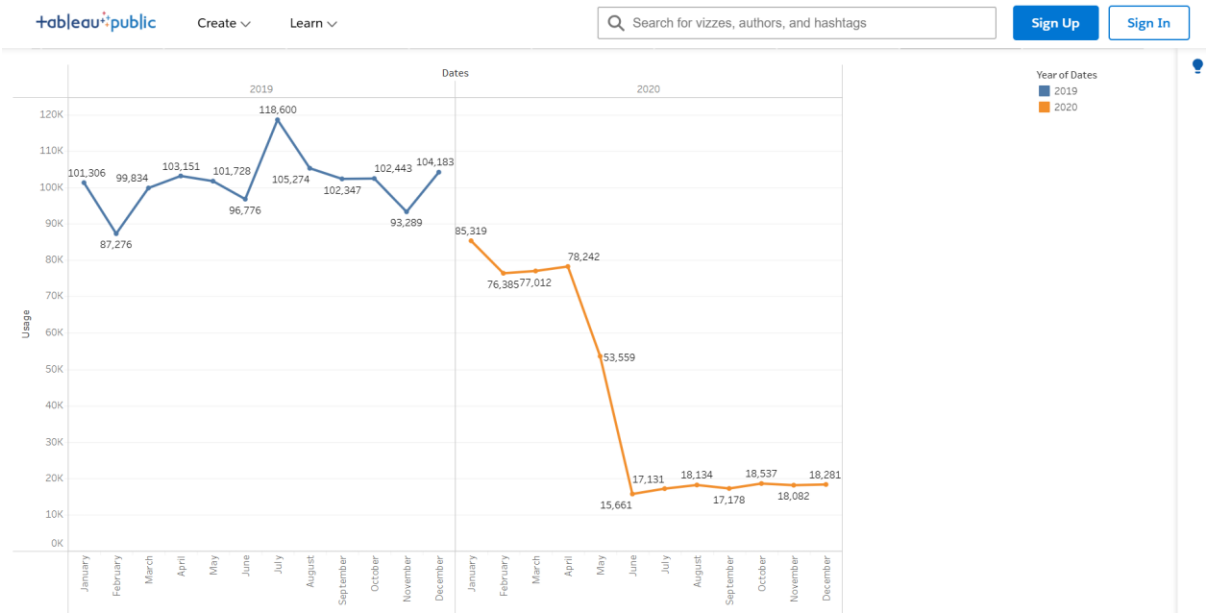
2) Quarter Usage :

- Drag **Date → Columns**.
- Click the Date field and change it to **Quarter**.
- Drag **Usage / Consumption → Rows**.
- Choose **Line Chart** or **Bar Chart**.



3) Usage by Year :

- Drag **Date → Columns**.
- Change Date format to **Year**.
- Drag **Usage / Consumption → Rows**.
- Select **Line Chart**.



5) Dashboard :

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖ **Creating Dashboard —**

Step 1: Create a New Dashboard :
a) At the bottom of Tableau, click the New Dashboard button (the icon next to the sheet tabs).
b) A blank dashboard canvas will open.
c) The dashboard will be named Dashboard 1 by default.

Step 2: Set the Dashboard Size :
a) On the left panel under Size, click the dropdown.
b) Select Fixed Size → choose Desktop Browser (1000 x 800) or select Automatic for responsive layout.

Step 3: Add Sheet 1 — Month-wise Consumption (2019 vs 2020) :
a) From the left Sheets panel, drag Sheet 1 (Month-wise Consumption 2019 vs 2020) onto the centre dashboard canvas.
b) Drop it in the top-left section of the dashboard layout.
c) On the top-right corner of the sheet, click the downward arrow (More Options).
d) Select Floating if you want to freely move the chart anywhere.
e) Or keep Tiled for automatic alignment in the dashboard.
f) Resize the sheet by dragging its edges so it fits properly in the top-left quadrant of the dashboard.

Step 4: Add Sheet 2 — Region Wise Usage :
a) From the Sheets panel, drag Sheet 2 (Region Wise Usage) onto the dashboard canvas.
b) Place it in the top-right section of the dashboard.
c) Click the downward arrow on the chart → choose Floating or Tiled layout depending on the design.
d) Adjust the width and height of the sheet so it fits neatly beside the first chart.

Step 5: Add Sheet 3 — State Wise Usage :
a) Drag Sheet 3 (State Wise Usage) from the Sheets panel to the dashboard.
b) Place it in the bottom-left section of the dashboard.
c) Use the drop-down arrow on the sheet to configure layout options if required.
d) Resize the chart so that it occupies the bottom-left quadrant.

Step 6: Add Sheet 4 — Quarter Wise Usage :

a) Drag Sheet 4 (Quarter Wise Usage) from the Sheets panel to the dashboard.
b) Drop it in the bottom-right area of the dashboard canvas.
c) Choose Floating or Tiled layout from the sheet options if necessary.
d) Adjust the chart size so that it fits the remaining space of the dashboard.

Step 6: Add Sheet 5 — Usage by Year :

a) Drag Sheet 5 (Year-wise Usage) from the Sheets panel.
b) Drop it in the third row, left side of the dashboard.
c) Configure Floating or Tiled layout if needed.
d) Adjust the chart size so it aligns with other sheets.

Step 7: Add Sheet 6 — Top N States by Consumption :

a) Drag Sheet 6 (Top States Consumption) onto the dashboard.
b) Place it in the third row, right side.
c) Use the drop-down menu on the sheet for layout settings.
d) Resize the chart so it fits properly in the dashboard section.

Step 8: Add Sheet 7 — Bottom N States by Consumption :

a) Drag Sheet 7 (Bottom States Consumption) from the Sheets panel.
b) Drop it in the fourth row, left section of the dashboard.
c) Choose Floating or Tiled depending on your design.
d) Adjust the chart size.

Step 9: Add Sheet 8 — Usage Before and After Lockdown :

a) Drag Sheet 8 (Before vs After Lockdown) to the dashboard.
b) Place it in the fourth row, right section.
c) Configure the layout using the sheet options arrow.
d) Resize the sheet to match the dashboard layout.

Step 10: Add Sheet 9 — Region and State Filter Panel :

a) Drag Sheet 9 (Filter Panel / Map / Summary Sheet) from the Sheets panel.
b) Place it at the top or side of the dashboard for easy filtering.
c) Enable Use as Filter from the sheet options.
d) Adjust the size so users can easily interact with it.

Step 11: Add a Dashboard Title :

a) At the bottom-left of the dashboard panel, check the box Show Dashboard Title.
b) Double-click the title area that appears at the top of the dashboard.
c) Type — Electricity Consumption Analysis Dashboard.
d) Set the font style to Bold, size 16, and color White.
e) Set the background color by clicking Format → Dashboard → Title background and select a suitable color such as Dark Blue or Red (#C0392B).

Step 12: Add Interactive Filters :

a) Click on any sheet inside the dashboard (for example Region Wise Usage).
b) Click the downward arrow on the top-right corner of that sheet.
c) Select Filters → check Region and State filters.
d) The filter widgets will appear on the right side of the dashboard.
e) Repeat the process for another sheet (for example Year Wise Usage) and add the Year filter.
f) These filters will now control all connected sheets simultaneously

Step 13: Format the Dashboard Background:

a) Click Format from the top menu.

b) Select Dashboard from the Format panel.

c) Under Background, set the color to White or light grey (#F4F4F4).

d) Under Dashboard Title, adjust the font size and background color.

e) To add borders to individual sheets:

- Click each sheet
- Select Format → Border
- Add an outer border in grey or dark blue.

Step 11: Publish the Dashboard to Tableau Public :

a) Click **File** from the **top menu** and select **Save to Tableau Public**.

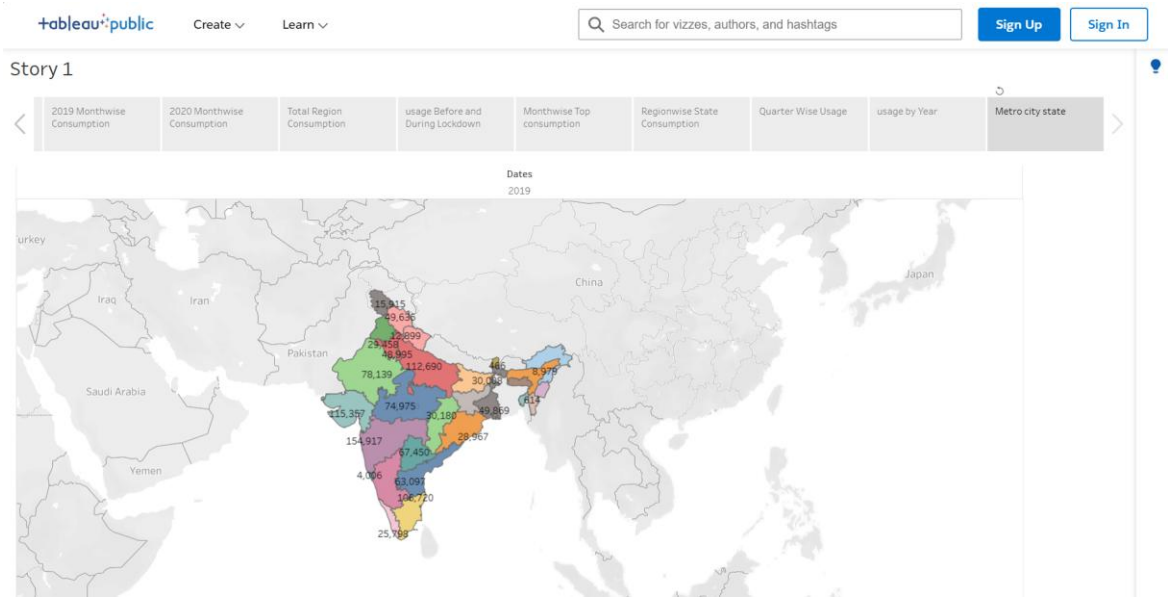
b) Sign in using your **Tableau Public account credentials** if prompted.

c) Name the workbook — **Electricity_Consumption_Dashboard** (or keep the existing name).

d) Click **Save**. Tableau will **upload the dashboard and open the live link in your browser**.

e) Your Dashboard 1 is now live at:

https://public.tableau.com/app/profile/tanuja.k
handare/viz/Book4_17728921597520/Story1



6) Storyboard :

A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

Step 1: Create New Story
a) At the bottom of Tableau, click the New Story button next to the sheet tabs.
b) A blank story canvas will open named Story 1.

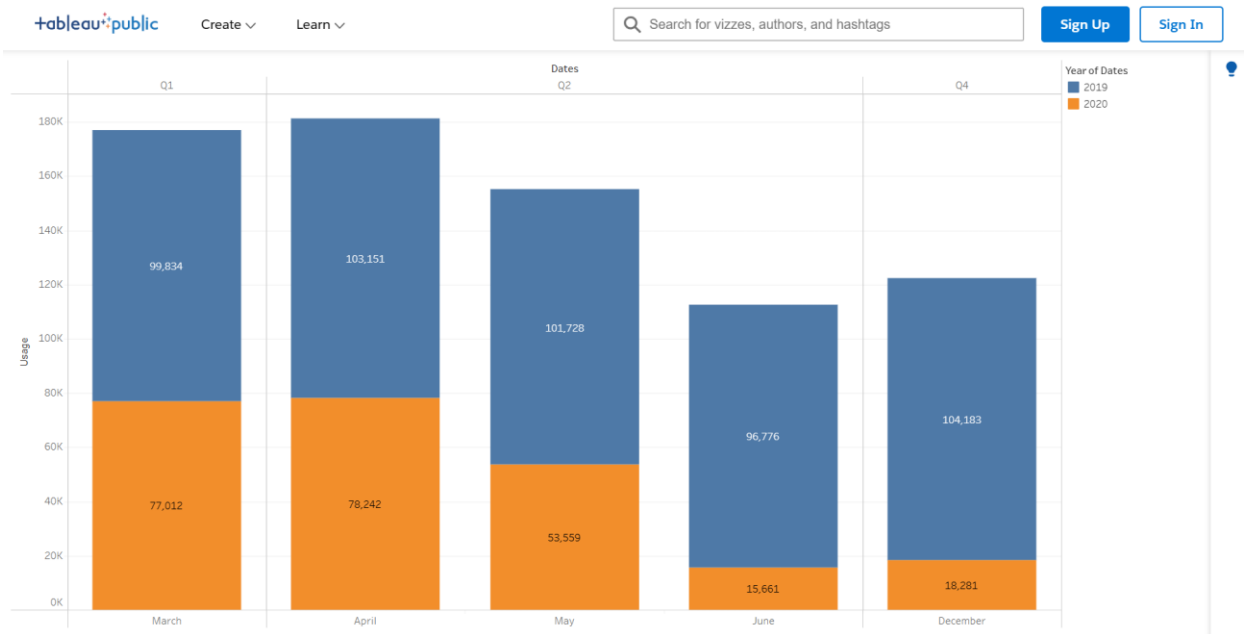
Step 2: Set Size
a) On the left panel under Size → select Automatic or Desktop Browser (1016 x 964).

Step 3: Add 10 Story Points
a) Click Blank on the left panel to create each new story point.
b) Drag and drop each sheet from the left Sheets panel onto the canvas one by one.
c) After dropping each sheet, click Add a Caption at the top and type the caption name.

Step 4: Add Story Title
a) Check Show Title at the bottom-left panel.
b) Double-click the title and type — Story.

Step 5: Publish
a) Click File → Save to Tableau Public.
b) Sign in → Name the workbook → Click Save.
c) Story goes live on your Tableau Public profile.

https://public.tableau.com/app/profile/tanuja.khandare/viz/Book4_17728921597520/Usagebeforeandduringlockdown?publish=yes



7) Web integration :

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

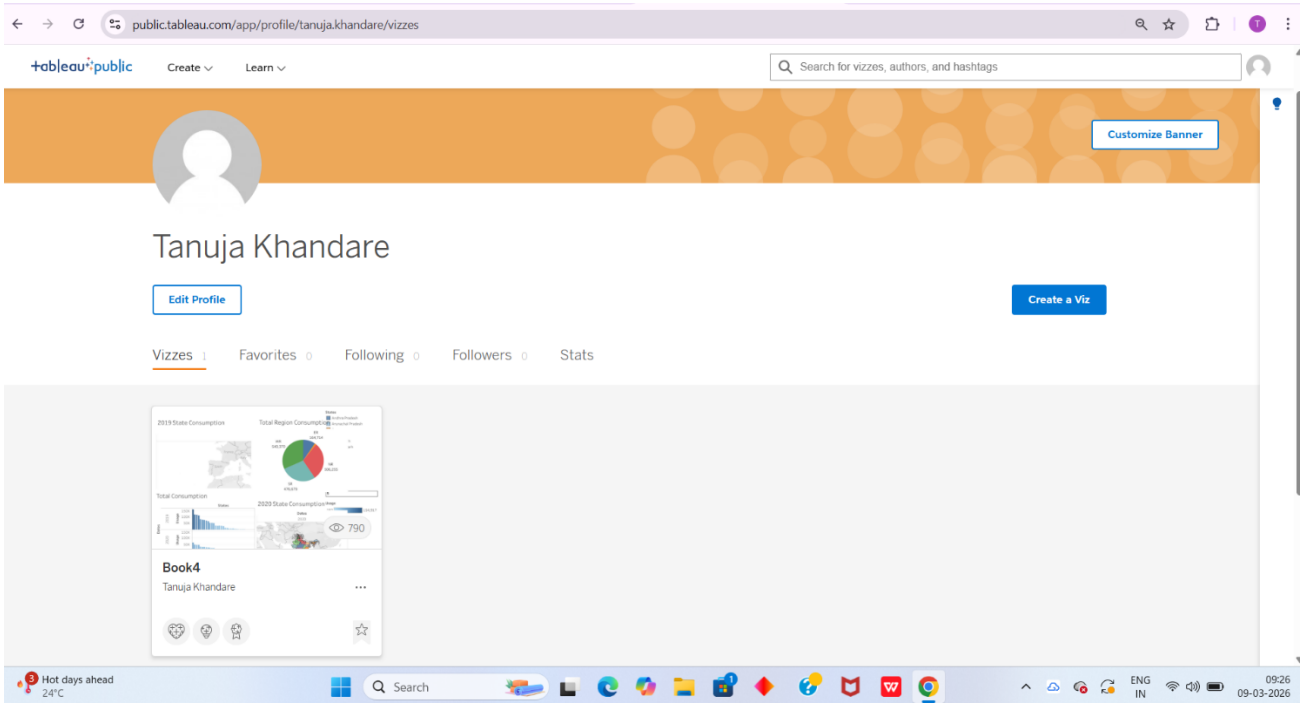
❖ Publishing

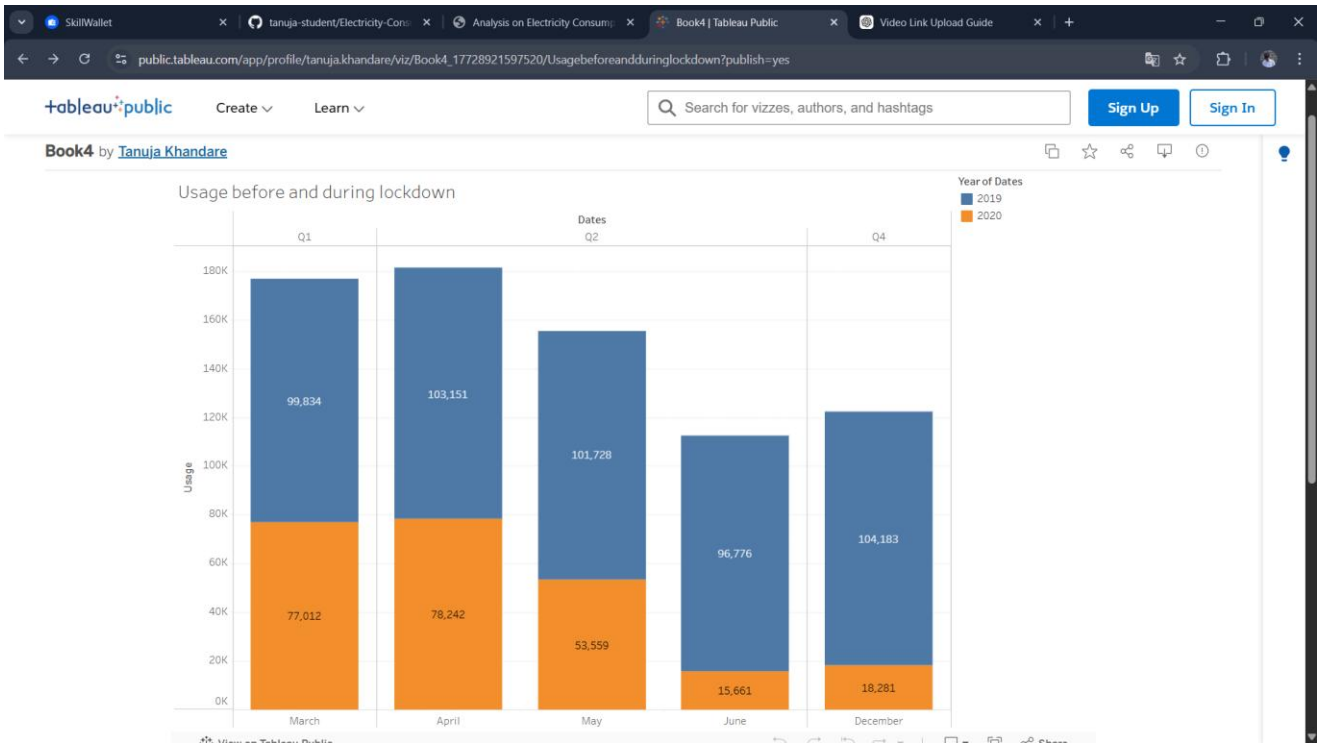
a) **Sign in to Tableau Public**

In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.

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Video Link Upload Guide
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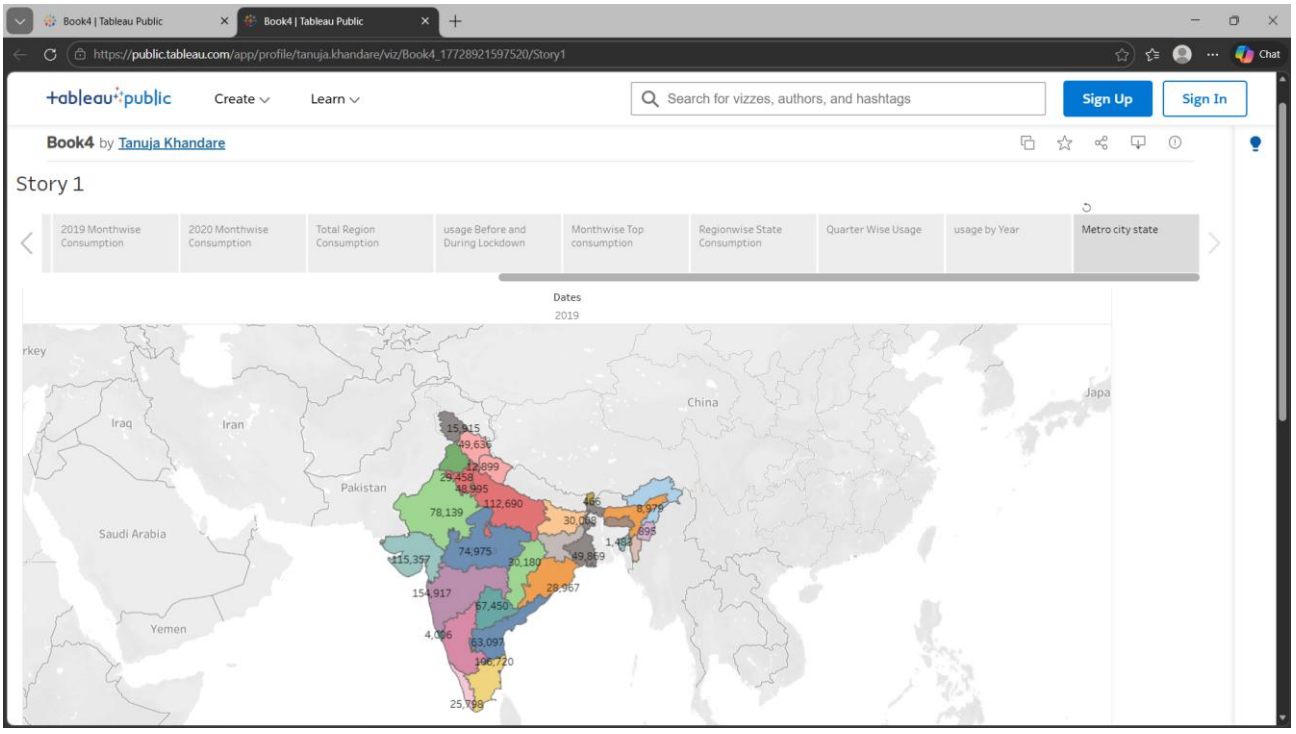
tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Sign Up
Sign In
Book4 by Tanuja Khandare
Usage before and during lockdown
Dates
Q1
Q2
Q4
Year of Dates
2019
2020
Usage
180K
160K
140K
120K
100K
80K
60K
40K
20K
0K
99,834
103,151
101,728
96,776
104,183
77,012
78,242
53,559
15,661
18,281
March
April
May
June
December


Book4 | Tableau Public
Book4 | Tableau Public
https://public.tableau.com/app/profile/tanuja.khandare/viz/Book4_17728921597520/Story1
Chat
tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Sign Up
Sign In
Book4 by Tanuja Khandare
Story 1
2019 Monthwise Consumption
2020 Monthwise Consumption
Total Region Consumption
usage Before and During Lockdown
Monthwise Top consumption
Regionwise State Consumption
Quarter Wise Usage
usage by Year
Metro city state
Dates
2019
China
Iraq
Iran
Pakistan
Saudi Arabia
Yemen
15,815
49,636
2,899
29,458
48,905
112,690
78,139
466
8,978
30,049
695
1,497
115,357
74,975
30,180
49,869
28,957
154,917
57,450
4,006
63,097
106,720
25,799

❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) app.py

app.py

C: > Users > LENOVO > OneDrive > Desktop > project > app.py

```python
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template("home.html")

@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html")

@app.route("/story")
def story():
    return render_template("story.html")

@app.route("/visualization")
def visualization():
    return render_template("visualization.html")

@app.route("/conclusion")
def conclusion():
    return render_template("conclusion.html")

if __name__ == "__main__":
    app.run(debug=True)
```

2) Adding Dashboard and Story on HTML page.

a) Open Tableau Public profile and open Dashboard/Story.

b) Click on share option and copy the embed code.

c) Paste in HTML page

app.py | story.html | home.html | visualization.html | dashboard.html | conclusion.html

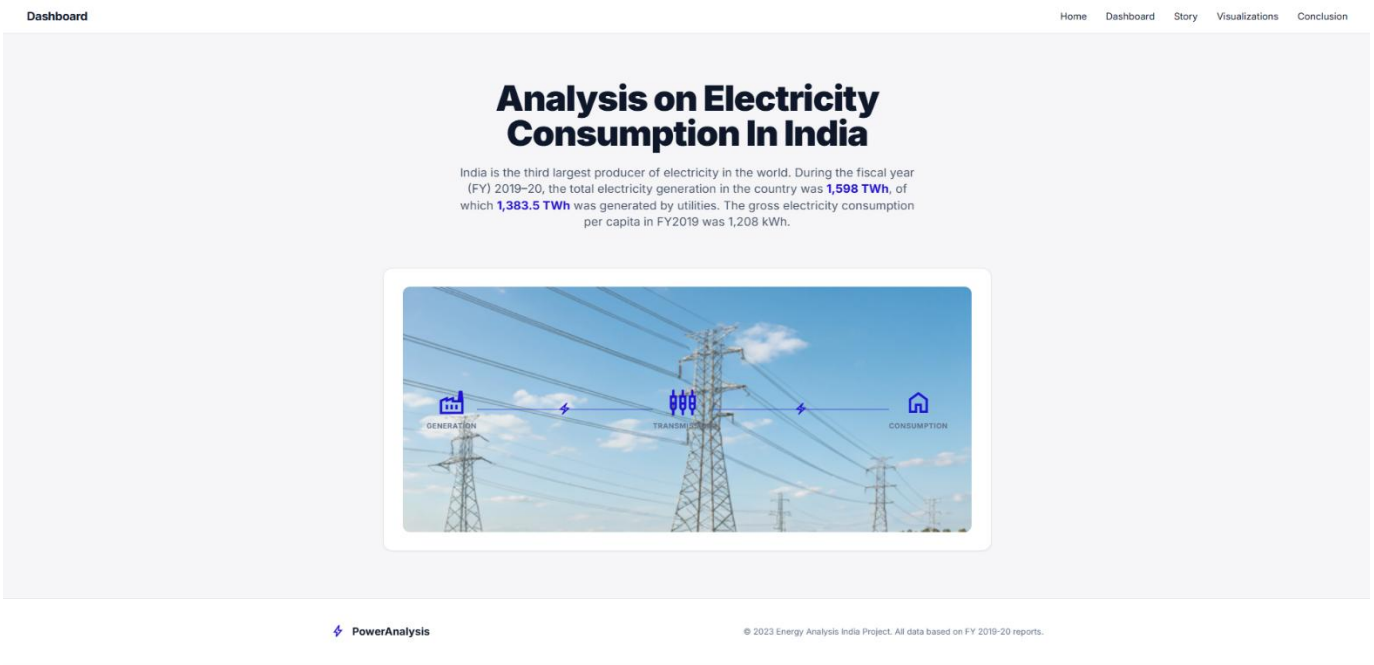
p.px-6.md:px-10.py-12.md:py-20.flex.flex-col.items-center.text-center > div.max-w-4xl.space-y-6 > p.text-slate-600.dark:text-slate-400.text-lg.md:text-xl.font-normal.leading-relaxed.

```html
<!DOCTYPE html>

<html lang="en">

<head>
    <meta charset="utf-8" />
    <meta content="width=device-width, initial-scale=1.0" name="viewport" />
    <title>Analysis on Electricity Consumption In India</title>
    <script src="https://cdn.tailwindcss.com?plugins=forms,container-queries"></script>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;700;900&amp;display=swap"
        rel="stylesheet" />
    <link href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:wght@100..700,0..1&amp;display=swap"
        rel="stylesheet" />
    <link
        href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:wght,FILL@100..700,0..1&amp;display=swap"
        rel="stylesheet" />
    <script id="tailwind-config">
        tailwind.config = {
            darkMode: "class",
            theme: {
                extend: {
                    colors: {
                        "primary": "#2111d4",
                        "background-light": "#f6f6f8",
                        "background-dark": "#121022",
                    },
                    fontFamily: {
                        "display": ["Inter", "sans-serif"]
                    },
                    borderRadius: {
                        "DEFAULT": "0.25rem",
                        "lg": "0.5rem",
                        "xl": "0.75rem",
                        "full": "9999px"
                    },
                },
            },
        }
    </script>
```

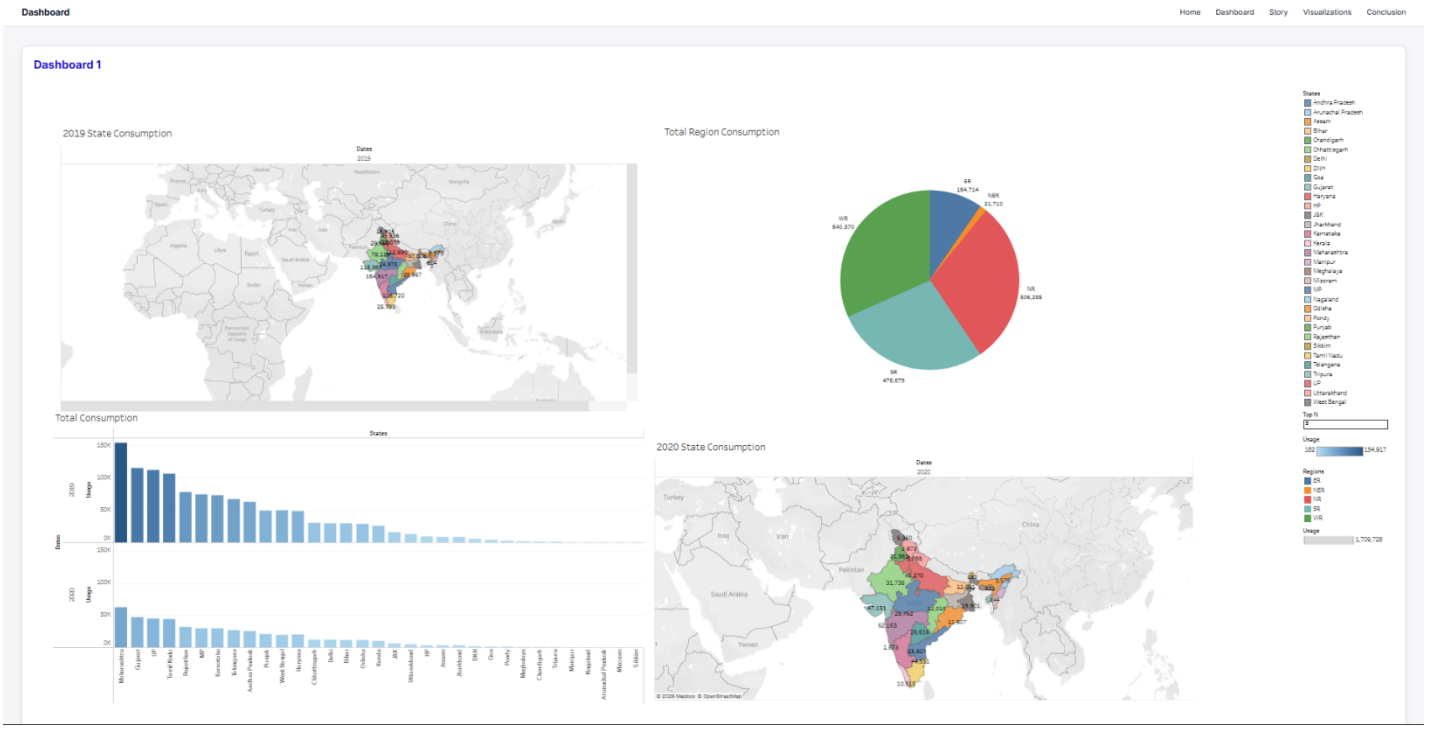
d) Run the app.py and copy the port number and paste it on Browser.

```
PS C:\Users\LENOVO\OneDrive\Desktop\project> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 644-262-450
```

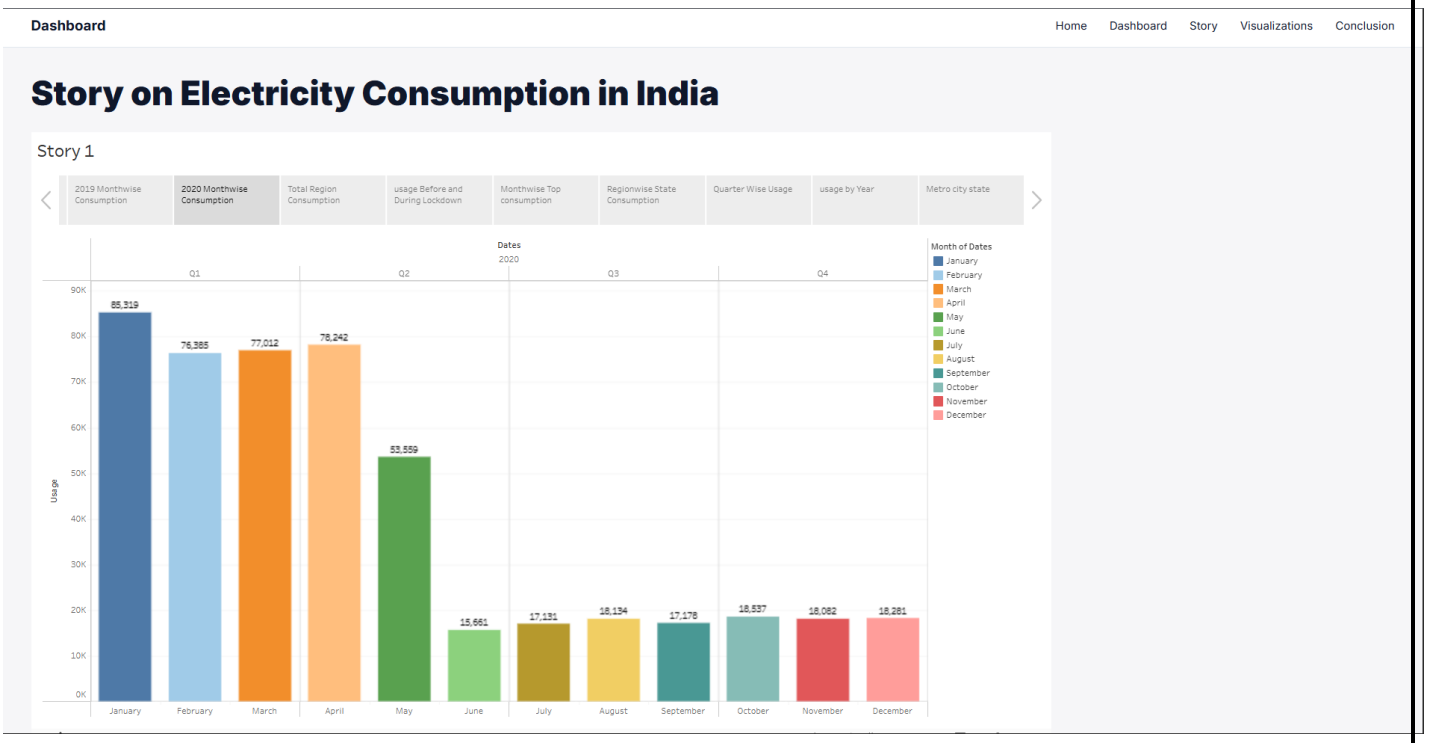
- **HOME :**



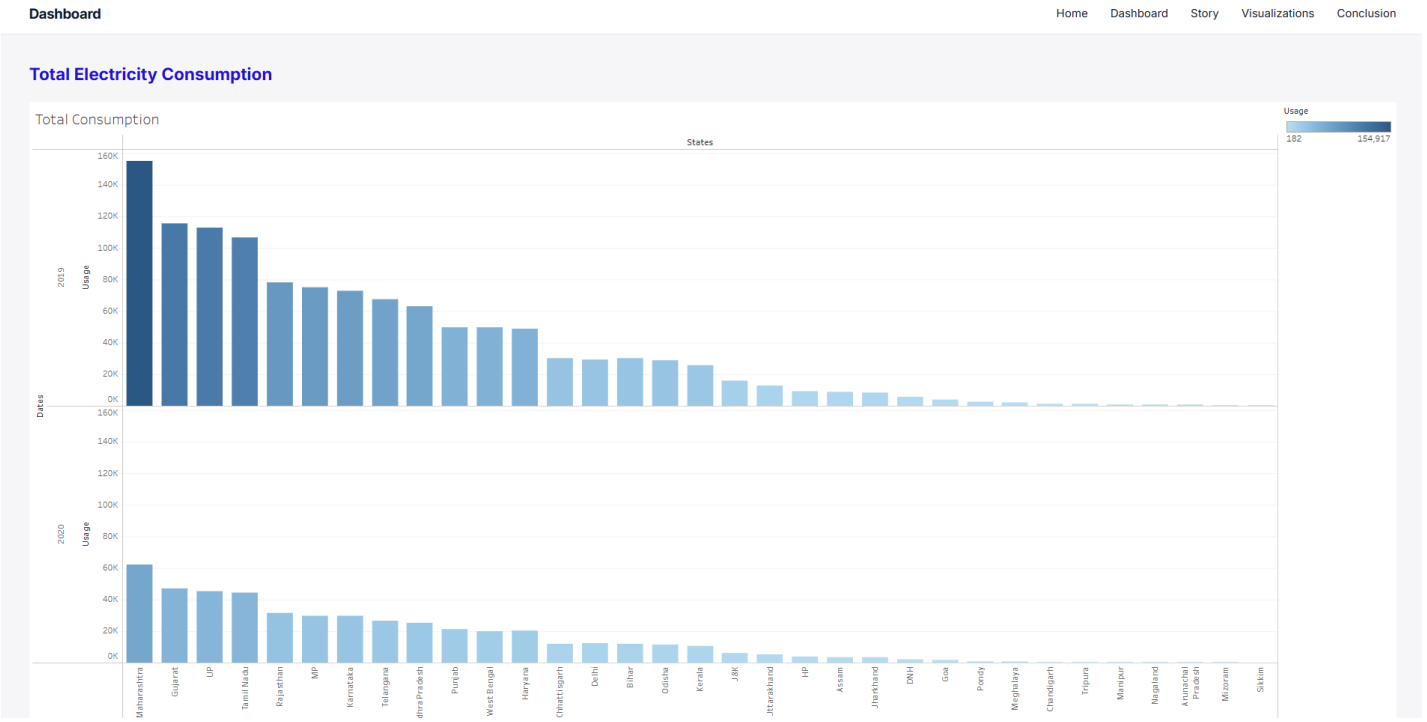
- **DASHBOARD :**



- **STORY :**



- **VISUALIZATION :**



- **CONCLUSION :**

Dashboard
Home
Dashboard
Story
Visualizations
Conclusion
Conclusion
Usage before and during lockdown
Year of Dates
2019
2020
Dates
Q1
Q2
Q4
Usage
150K
100K
50K
0K
99,834
77,012
103,151
78,242
101,728
53,559
96,776
104,183
18,281
March
April
May
June
December
View on Tableau Public
Share
Electricity Consumption Stats
✓ Maharashtra is the Highest Electricity consumption user of India.
✓ Gujarat is the Second Highest Electricity consumption user of India.
✓ Sikkim is the Lowest Electricity Consumption user of India.
© 2024 India Electricity Analysis. All data based on FY 2019-20 reports.